Excitations of 1d spin chain

Ising limit $H = \sum_n \underbrace{J_z S_n^z S_{n+1}^z}_{H_z} + \underbrace{\frac{1}{2} J_\perp (S_n^+ S_{n+1}^- + S_n^- S_{n+1}^+)}_{H_\perp}$

$J_z \gg J_\perp > 0$.

lowest energy state $|A\rangle = |\overset{1}{\uparrow}\overset{2}{\downarrow}\overset{3}{\uparrow}\overset{4}{\downarrow}\overset{}{\uparrow}\overset{N}{\downarrow}\rangle$ say spin-$\uparrow$ on A sublattice

$|B\rangle = |\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\rangle$ spin-$\uparrow$ on B sublat.

Domain wall $|\psi_n\rangle = \uparrow\downarrow\overset{n-1}{\uparrow}\overset{n}{\downarrow}\,|\,\overset{n+1}{\downarrow}\uparrow\downarrow\uparrow$

$\underbrace{\qquad\qquad}_{A}\,|\,\underbrace{\qquad\qquad}_{B}$

↳ "ferromagnetic" bond between sites $\underline{n}$ and $\underline{n+1}$

$$H_z |A\rangle = J_z \sum_n \frac{1}{2}\left(-\frac{1}{2}\right) = -\frac{1}{4} J_z \cdot N$$

$$H_z |B\rangle = -\frac{1}{4} J_z \cdot N$$

$$H_z |\psi_n\rangle = J_z \Big\{ \sum_{n=1}^{n} \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \sum_{n=n+1}^{N} \left(-\frac{1}{2}\right)\cdot\frac{1}{2} + \underbrace{\left(-\frac{1}{2}\right)\cdot\left(-\frac{1}{2}\right)}_{\text{bond } (n,n+1)} \Big\} = -\frac{1}{4} J_z \left(n + N - (n+1)\right) + \frac{1}{4} J_z$$

$$= -\frac{1}{4} J_z N + \frac{1}{2} J_z$$

Observe that single-spin flip state $\uparrow\downarrow\uparrow\downarrow\downarrow\downarrow\uparrow\downarrow\uparrow$

$\quad\;\; 1\;\;2\;\;3\;\;\underline{4\;\;5}\;\;6\;\;7\;\;8\;\;9$

is a state with 2 domain walls, on bonds (4,5) and (5,6).

$\quad\;\; 1\;\;2\;\;3\;\;4\;\;5\;\;6\;\;7\;\;8\;\;9$

$\quad\;\; \uparrow\downarrow\uparrow\downarrow\downarrow\uparrow\downarrow\downarrow\uparrow$

$\quad\;\; \underbrace{\qquad}_{A}\;\;\underbrace{\qquad}_{B}\;\;\underbrace{\qquad}_{A}$

Simple classical stat. mech — every bond of the 1d chain can be in one of the two states: $\begin{cases} AF & \uparrow\downarrow, \downarrow\uparrow \\ F & \uparrow\uparrow, \downarrow\downarrow \end{cases}$

$$\varepsilon_{AF} = -\frac{J_z}{4}, \quad \varepsilon_F = \frac{J_z}{4}$$

$\Rightarrow$ bonds partition function $Z_B = e^{-\varepsilon_{AF}/T} + e^{\varepsilon_F/T} =$

$$= 2\cosh\left(\frac{J_z}{4T}\right)$$

For a N-bond-long chain $Z = Z_B^N$

<u>Average!</u> Number of domain walls = number of F-bonds =

$$\overline{N} = P_F \cdot N = \frac{e^{-J_z/4T}\, N}{2\cosh(J_z/4T)} = \frac{N e^{-J_z/4T}}{e^{J_z/4T} + e^{-J_z/4T}} =$$

$$= \frac{N}{e^{J_z/2T} + 1}$$

Note that domain walls are like fermions, as only one can be present on a given link.

Consider now spin-correlation function $\langle S_0^z S_n^z \rangle$. In a perfectly ordered state

$$\langle S_0^z S_n^z \rangle = \frac{(-1)^n}{4} = \begin{cases} \frac{1}{2}\cdot\frac{1}{2}, & n = \text{even} \\ \frac{1}{2}\cdot\left(-\frac{1}{2}\right) = \left(-\frac{1}{2}\right)\cdot\frac{1}{2} & \text{for } n = \text{odd} \end{cases}$$

	1	2	3	4	5	6	7	8	9	10
ideal state	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$
1 domain wall	$\uparrow$	$\downarrow$	$\uparrow$	$\underline{\underline{\uparrow}}$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$
2 walls	$\uparrow$	$\downarrow$	$\uparrow$	$\underline{\underline{\uparrow}}$	$\downarrow$	$\uparrow$	$\underline{\underline{\downarrow}}$	$\downarrow$	$\uparrow$	$\downarrow$
$\vdots$										

However, if there are N domain walls in between spins S_0^z and S_n^z, then

$$\langle S_0^z S_n^z \rangle = \frac{(-1)^{n+N}}{4}$$

Thus we need to calculate

$$4\langle S_0^z S_n^z \rangle = \langle (-1)^{n+N} \rangle$$

← means average over all possible states of the segment of length n between sites 0 and n.

Caution: $\langle (-1)^N \rangle \neq (-1)^{\langle N \rangle}$

In fact: $$\langle (-1)^N \rangle = \frac{\sum_{N=0}^{n} (-1)^N p_N}{\sum_{N=0}^{n} p_N}$$

where p_N = probability to have N domain walls in a segment of length n

let Energy of single domain wall <u>relative</u> to the energy of ideal state be $\Delta = J_z/2$.

Then $$p_N = e^{-N\Delta/T} \frac{n!}{N!\,(n-N)!}$$

Boltzmann probability of N-domain state

number of ways to arrange N objects amongst n bonds

Then $$\sum_{N=0}^{n} p_N = \sum_{N=0}^{n} \frac{n!}{N!\,(n-N)!} \left(e^{-\Delta/T}\right)^N = \left(1 + e^{-\Delta/T}\right)^n$$

binomial expansion!

$$\sum_{N=0}^{n} (-1)^N p_N = \sum_{N=0}^{n} \frac{n!}{N!(n-N)!} \left(-e^{-\Delta/T}\right)^N =$$

$$= \left(1 - e^{-\Delta/T}\right)^n$$

$$\Rightarrow \langle (-1)^N \rangle = \frac{\left(1 - e^{-\Delta/T}\right)^n}{\left(1 + e^{-\Delta/T}\right)^n} = \left(\frac{2\,\mathrm{sh}\,\frac{\Delta}{2T}}{2\,\mathrm{ch}\,\frac{\Delta}{2T}}\right)^n =$$

$$= \left(\tanh \frac{J_z}{4T}\right)^n.$$

$$\Rightarrow 4\langle S_0^z S_n^z \rangle = (-1)^n \left(\tanh \frac{J_z}{4T}\right)^n.$$

$$4\langle S_0^z S_n^z\rangle = \left\{-\tanh\frac{J_z}{4T}\right\}^n = (-1)^n\, e^{n\ln\left[\tanh\left(\frac{J_z}{4T}\right)\right]}$$

$$= (-1)^n\, e^{-n/\xi}$$

$$\xi = -\frac{1}{\ln\left[\tanh\left(\frac{J_z}{4T}\right)\right]} = \frac{1}{\ln\left[\frac{\cosh(J_z/4T)}{\sinh(J_z/4T)}\right]} =$$

$$= \begin{cases} T \gg J_z, & \dfrac{-1}{\ln(J_z/4T)} = \dfrac{1}{\ln\left(\frac{4T}{J_z}\right)} \\ T \ll J_z, & 1/\left(2e^{-J_z/4T}\right) = \frac{1}{2}e^{J_z/4T} \end{cases}$$

$$\ln\left(\frac{\cosh x}{\sinh x}\right) \xrightarrow[x\to\infty]{} \ln\left(\frac{e^{x/2}+e^{-x/2}}{e^{x/2}-e^{-x/2}}\right) = \ln\left(\frac{1+e^{-x}}{1-e^{-x}}\right) =$$

$$\simeq \ln\left(1+2e^{-x}\right) = 2e^{-x}$$

Thus spin-correlation length $\xi \sim e^{J_z/4T} \to \infty$

as $T \to 0 \Rightarrow$ spin system ~~approach~~ approaches ordered state.

But at any $T \neq 0$ $\xi(T)$ is finite

$$\Rightarrow \langle S_0^z S_n^z\rangle \sim e^{-n/\xi} \xrightarrow[n\to\infty]{} 0.$$

Now quantum problem: $\frac{J_\perp}{J_z} \neq 0$

$$\mathcal{H}_\perp = \frac{1}{2} J_\perp \sum_n \left(S_n^+ S_{n+1}^- + S_n^- S_{n+1}^+ \right)$$

$\mathcal{H}_\perp |A\rangle$ is a state with 2 domain walls

$\Rightarrow$ has an energy $2 \cdot \left(\frac{1}{2} J_z\right)$ relative to E_A.

Same for $\mathcal{H}_\perp |B\rangle$ $\rightarrow$ focus on a <u>single kink</u> state $|\psi_n\rangle$

$$\mathcal{H}_\perp |\psi_n\rangle \cong \frac{J_\perp}{2} \left(S_{n-1}^- S_n^+ + S_{n+1}^+ S_{n+2}^- \right) |\uparrow\downarrow\overset{n-1}{\uparrow}\overset{n}{\downarrow}\times\downarrow\overset{n+1}{\uparrow}\overset{n+2}{\downarrow}\rangle$$

$$= \frac{J_\perp}{2} \left\{ |\uparrow\overset{n-2}{\downarrow}\overset{n-1}{\times}\downarrow\overset{n}{\uparrow}\overset{n+1}{\downarrow}\overset{n+2}{\uparrow}\rangle + \right.$$

$$\left. + |\uparrow\downarrow\overset{n-1}{\uparrow}\overset{n}{\downarrow}\overset{n+1}{\uparrow}\overset{n+2}{\downarrow}\times\downarrow\rangle \right\} =$$

$$= \frac{J_\perp}{2} \left(|\psi_{n-2}\rangle + |\psi_{n+2}\rangle \right)$$

$$\Rightarrow \mathcal{H} |\psi_n\rangle = \left(\mathcal{H}_z + \mathcal{H}_\perp\right) |\psi_n\rangle = \frac{J_z}{2} |\psi_n\rangle + \frac{J_\perp}{2} \left(|\psi_{n-2}\rangle + |\psi_{n+2}\rangle \right)$$

within <u>1-kink</u> (1 domain wall) subspace.

$\Rightarrow$ plane-wave state $|1k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{ikn} |\psi_n\rangle$

is approximate eigenstate ~~too~~:

$$\mathcal{H} |1k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{ikn} \mathcal{H} |\psi_n\rangle =$$

$$= \frac{1}{\sqrt{N}} \sum_n e^{ikn} \left(\frac{J_z}{2} |\psi_n\rangle + \frac{J_\perp}{2} \left(|\psi_{n+2}\rangle + |\psi_{n-2}\rangle \right) \right) =$$

$$= \frac{J_z}{2} |1k\rangle + \frac{J_\perp}{2} \left(e^{-i2k} + e^{i2k} \right) |1k\rangle =$$

$$= \left[\frac{J_z}{2} + J_\perp \cos(2k) \right] |1k\rangle$$

$\Rightarrow$ energy of 1-kink state relative to the ground state

$$\mathcal{E}_1(k) = \frac{J_z}{2} + J_\perp \cos(2k)$$

Now, in a system with periodic boundary conditions (PBC) only even number of kinks is allowed

$\Rightarrow$ first excited state involves <u>2 kinks</u>.

We are allowed to treat <u>the kinks</u> as non-interacting particles (try to check this yourselves)

$\Rightarrow$ 2-kink state with total energy E and momentum q must be a continuum of $|k_1\rangle$ and $|k_2\rangle$

$$\begin{cases} q = k_1 + k_2 \\ E = \mathcal{E}(k_1) + \mathcal{E}(k_2) \end{cases}$$

$$\cos 2k_1 + \cos 2k_2 = 2\cos(k_1+k_2)\cos(k_1-k_2)$$

$$\frac{1}{2}\left(e^{i(k_1+k_2)} + e^{-i(k_1+k_2)}\right)\left(e^{i(k_1-k_2)} + e^{-i(k_1-k_2)}\right)$$

$$= \frac{1}{2}\left(e^{i2k_1} + e^{i2k_2} + e^{-i2k_2} + e^{-i2k_1}\right)$$

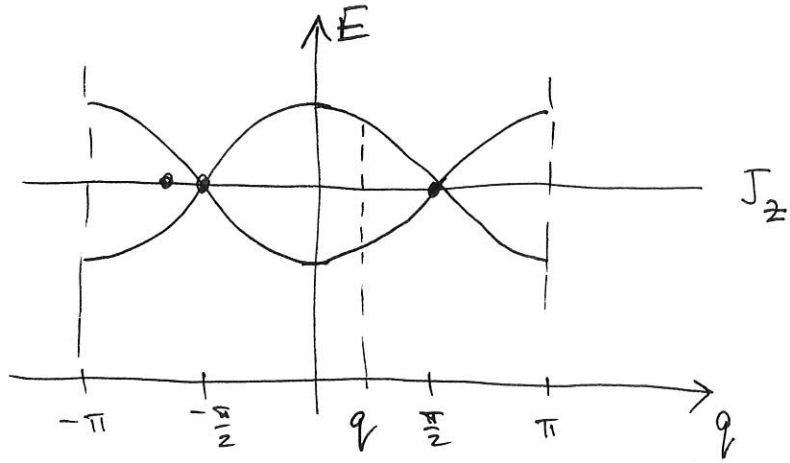
$$E = J_z + 2J_\perp \cos(q)\cos(q-2k_2)$$

$$\cos(q-2k_2) = -1 \Rightarrow \quad q - 2k_2 = \pi$$

$$k_2 = \frac{1}{2}(q-\pi) = -\frac{\pi}{2} + \frac{q}{2}$$

$$k_1 = q - k_2 = \frac{1}{2}(q+\pi) = \frac{\pi}{2} + \frac{q}{2}$$

$$\cos(q-2k_2) = 1 \Rightarrow \quad q = 2k_2, \quad k_2 = \frac{1}{2}q \Rightarrow k_1 = \frac{1}{2}q$$



Key point: for a given momentum q the energy of 2-kink state can take values from the band of width $4J_\perp\cos(q)$ centered around J_z.

Note $k_{1,2} = \pm\frac{\pi}{2} + \frac{q}{2} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

while $q \in (-\pi, \pi)$

For a given q: $\Delta E = J_z + 2J_\perp \cos(q)\cos(q-2k_2)$

$$= J_z + 2J_\perp\cos(q)\begin{cases} +1 & \text{upper} \\ -1 & \text{lower} \end{cases} \text{boundaries}$$

Heisenberg chain and Jordan-Wigner transformation:

we would like to understand $J_\perp \to J_z$ limit, when

$$\mathcal{H} \to \mathcal{H}_{HEIS} = J \sum_n \vec{S}_n \cdot \vec{S}_{n+1}$$

For $\frac{1}{2}$-spins $\vec{S} = \frac{\hbar}{2} \vec{\sigma} = \frac{\hbar}{2}(\sigma^x, \sigma^y, \sigma^z)$

↳ Pauli matrices

For spins on the <u>same site</u>

$$[\sigma^a, \sigma^b] = 2i\,\varepsilon^{abc}\,\sigma^c \quad , \varepsilon^{abc} = \text{completely asymmetric tensor}$$

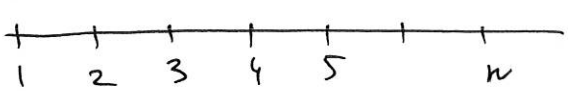
$$\sigma^a \sigma^b = i\,\varepsilon^{abc}\,\sigma^c \qquad \varepsilon^{123} = 1$$

While on different sites $[\sigma^a_n, \sigma^b_m] = 0$
they commute.

Consider now $\sigma^+_n = \left(e^{i\pi \sum_{j=1}^{n-1} a^\dagger_j a_j}\right) a^\dagger_n = K_n a^\dagger_n$

$(\sigma^\pm = \sigma^x \pm i\sigma^y)$ $\quad \sigma^-_n = a_n K_n \quad , \sigma^z_n = a^\dagger_n a_n - \frac{1}{2}$

string operator $K_n = K_n^\dagger = \exp\left(i\pi \sum_{j=1}^{n-1} a^\dagger_j a_j\right) =$



$$= \prod_{j=1}^{n-1} \left(e^{i\pi a^\dagger_j a_j}\right)$$

$\hat{a}_j$ is Fermion annihilation operator

$\hat{a}^\dagger_j \hat{a}_j$ is a number operator with eigenvalues 0, 1

$$\Rightarrow e^{i\pi a^\dagger_j a_j} = 1 - 2a^\dagger_j a_j = \begin{cases} 1, & \text{if } a^\dagger_j a_j |j\rangle = 0|j\rangle \\ -1, & \text{if } \hat{a}^\dagger_j \hat{a}_j |j\rangle = 1|j\rangle \end{cases}$$

$$= -2\sigma^z_j$$

Few useful observations: $a_n K_n = K_n a_n$ etc.

because K_n actually does not have operators from site n.

Also, $(K_n)^2 = 1$ because $e^{i2\pi \hat{a}_j^+ \hat{a}_j} = \hat{1}$.

Then $\sigma_n^+ \sigma_n^- = K_n a_n^+ a_n K_n = a_n^+ K_n K_n a_n = a_n^+ a_n$

$$\sigma_n^- \sigma_n^+ = a_n K_n K_n a_n^+ = a_n a_n^+ = 1 - a_n^+ a_n$$

$$\Rightarrow \sigma_n^+ \sigma_n^- - \sigma_n^- \sigma_n^+ = 2 a_n^+ a_n - 1 = 2\sigma_n^z.$$

Next, $\underline{m>n}$ $\sigma_n^+ \sigma_m^- = a_n^+ \underbrace{K_n K_m}_{e^{i\pi \sum_{j=n}^{m-1} a_j^+ a_j}} a_m = e^{i\pi \sum_{j=n+1}^{m-1} a_j^+ a_j} a_n^+ a_m$

$$\sigma_m^- \sigma_n^+ = a_m \overbrace{K_m K_n}^{e^{i\pi \sum_{j=n}^{m-1} a_j^+ a_j}} a_n^+ = K_m K_n a_m a_n^+ = -e^{i\pi \sum_{n+1}^{m-1} a_j^+ a_j} a_m a_n^+$$

$$\cancel{a_n^+ e^{i\pi a_n^+ a_n} = a_n^+ (1 - 2a_n^+ a_n) = a_n^+ - 2 \underbrace{a_n^+ a_n^+}_{0} a_n = a_n^+}$$

$$e^{i\pi a_n^+ a_n} a_n^+ = (1 - 2a_n^+ a_n) a_n^+ =$$

$$= a_n^+ - 2 a_n^+ \underbrace{a_n a_n^+}_{1 - a_n^+ a_n} =$$

$$= a_n^+ - 2a_n^+ + 2 \underbrace{a_n^+ a_n^+}_{0} a_n = -a_n^+$$

$$\Rightarrow [\sigma_n^+, \sigma_m^-] = e^{i\pi \sum_{n+1}^{m-1} a_j^+ a_j} (a_n^+ a_m - (-a_m a_n^+))$$

$$a_n^+ a_m + a_m a_n^+ = 0 \quad n \neq m$$

$\Rightarrow \sigma_n^+ \sigma_m^- = \sigma_m^- \sigma_n^+$ for $n \neq m$, as it should be.

Finally, $\mathcal{H}_\perp = \frac{1}{2} J_\perp \sum_n \left(S_n^+ S_{n+1}^- + S_n^- S_{n+1}^+ \right)$

$$= \frac{1}{2} J_\perp \sum_n \frac{1}{4} \left(a_n^+ K_n K_{n+1} a_{n+1} + h.c. \right)$$

$$K_n K_{n+1} = e^{i\pi \sum_{j=1}^{n-1} a_j^+ a_j} \, e^{i\pi \sum_{\ell=1}^{n} a_\ell^+ a_\ell} =$$

$$= e^{i 2\pi \sum_{j=1}^{n-1} a_j^+ a_j + i\pi a_n^+ a_n} = e^{i\pi a_n^+ a_n}$$

$$\Rightarrow a_n^+ K_n K_{n+1} a_{n+1} = a_n^+ e^{i\pi a_n^+ a_n} a_{n+1} =$$

$$= a_n^+ \left(1 - 2 a_n^+ a_n\right) a_{n+1} = \left(a_n^+ - 2 \underbrace{a_n^+ a_n^+}_{\emptyset} a_n\right) a_{n+1} =$$

$$= a_n^+ a_{n+1}$$

$$\Rightarrow \mathcal{H}_\perp = \frac{J_\perp}{8} \sum_n \left(a_n^+ a_{n+1} + a_{n+1}^+ a_n \right) \leftarrow \text{freely moving fermions}$$

$$\mathcal{H}_z = J_z \sum_n S_n^z S_{n+1}^z = \frac{J_z}{4} \sum_n \sigma_n^z \sigma_{n+1}^z =$$

$$= \frac{J_z}{4} \sum_n \left(a_n^+ a_n - \frac{1}{2} \right) \left(a_{n+1}^+ a_{n+1} - \frac{1}{2} \right)$$

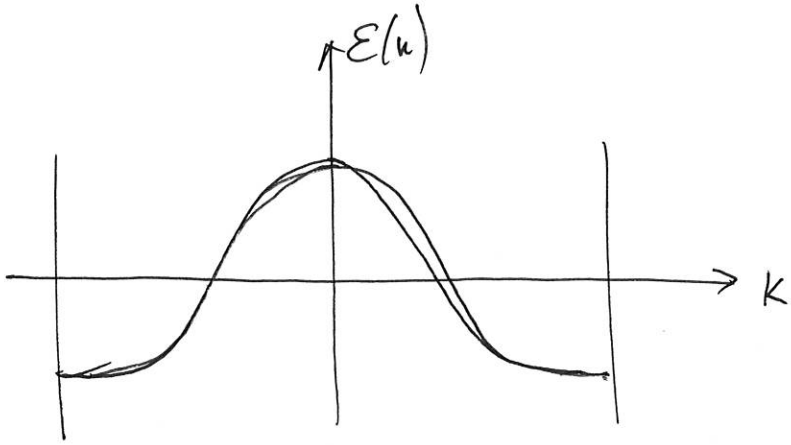
is interaction of fermion densities.

Thus problem of interacting spins is mapped onto that of interacting fermions.

But note that for $J_z = 0$ (XY model)

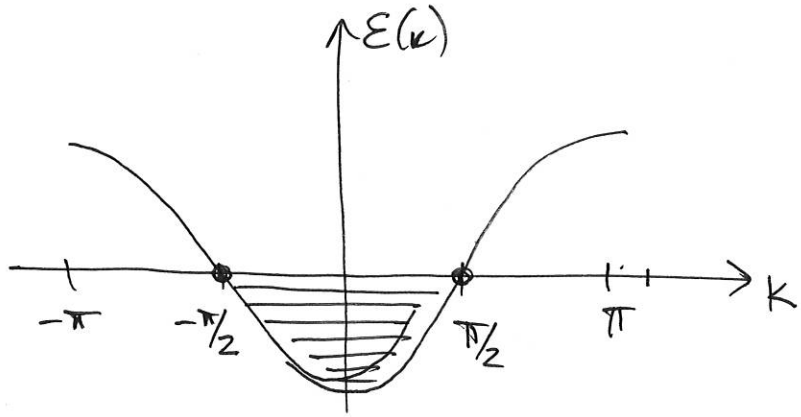
$$\mathcal{H}_\perp = \frac{J_\perp}{8} \sum_n \left(a_n^+ a_{n+1} + h.c. \right) = \sum_k \frac{J_\perp}{4} \cos k \, a_k^+ a_k$$

is Hamiltonian of free fermions!



Can just shift k by π: $k \to k+\pi$, so that

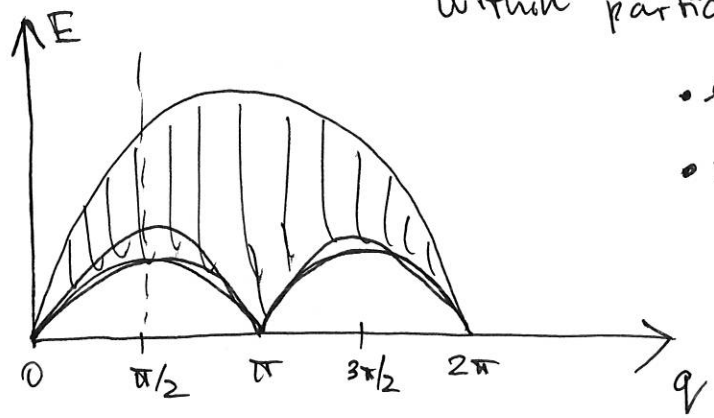
$$\mathcal{E}(k) \to -\frac{J_\perp}{4}\cos k$$



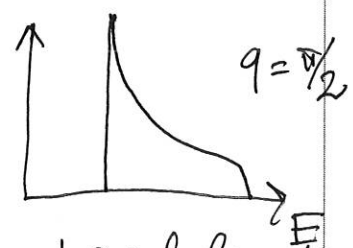
$\langle S^z_n \rangle = 0 \Rightarrow \langle a^\dagger_n a_n \rangle = \frac{1}{2} \Rightarrow$ the band is $\frac{1}{2}$-filled

$\Rightarrow k_F$ points are at $\pm\frac{\pi}{2}$.

$\Rightarrow$ dynamic spin correlations have support within particle-hole continuum



- extends to $E=0$
- most intensity near $q=\pi$ and lower boundary



- non-trivial lineshape